Project Design Phase-II

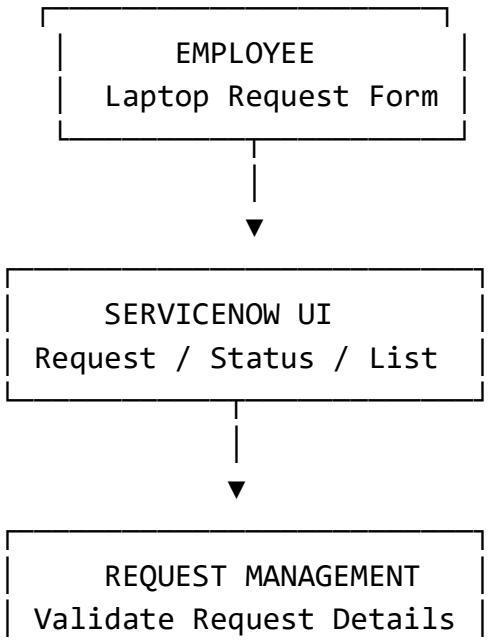
Technology Stack (Architecture & Stack)

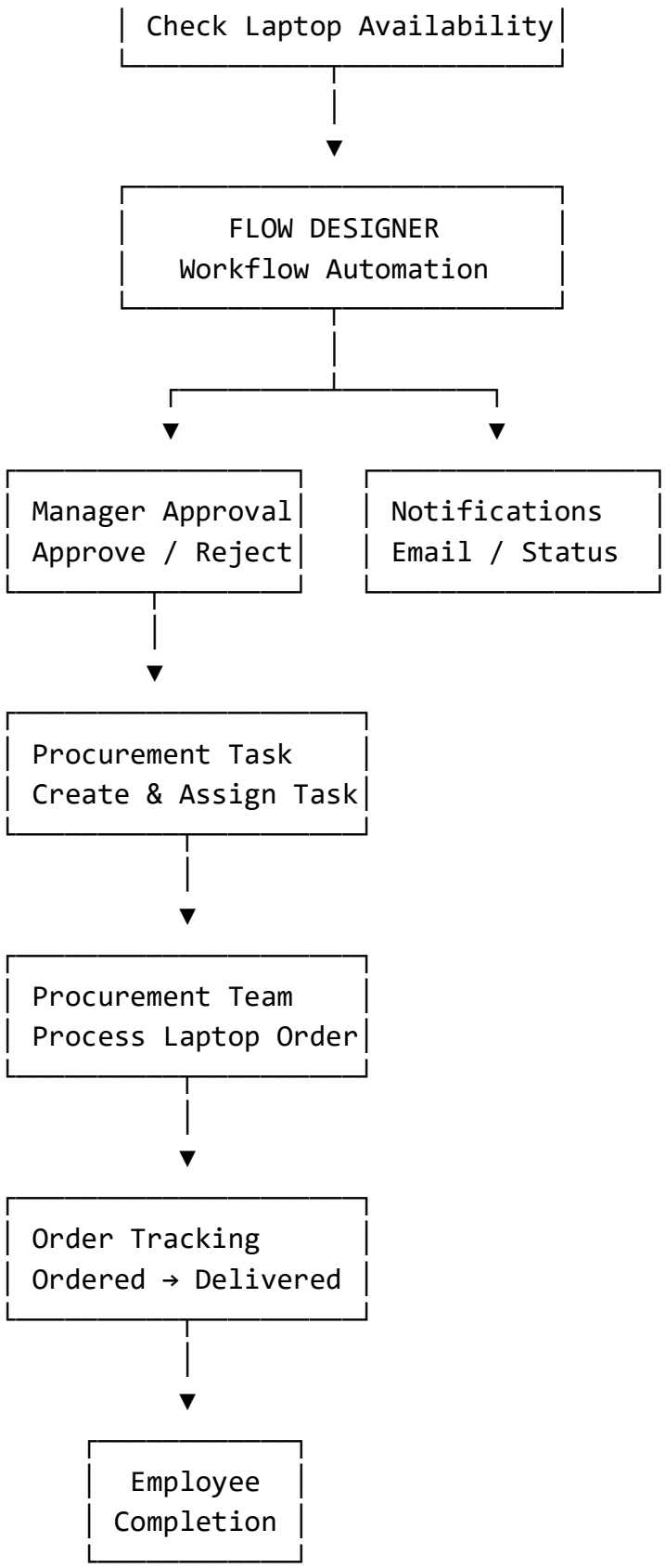
Date	03 October 2026
Team ID	SWTID-2026-5094
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

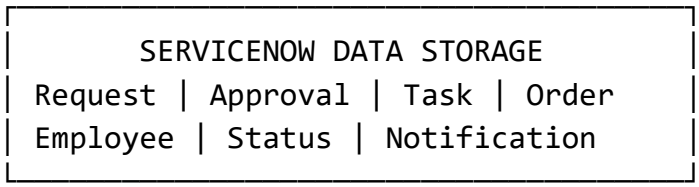
1. Technical Architecture

The proposed solution uses **ServiceNow** as the central platform. Employees submit laptop requests through the ServiceNow interface. The request is validated and processed through **Flow Designer**, which manages approval, procurement tasks, notifications, and order tracking.

Architecture Diagram




Check Laptop Availability
FLOW DESIGNER
Workflow Automation
Manager Approval
Approve / Reject
Notifications
Email / Status
Procurement Task
Create & Assign Task
Procurement Team
Process Laptop Order
Order Tracking
Ordered → Delivered
Employee
Completion



2. Table-1: Components & Technologies

S.No	Component	Description	Technology
1	**User Interface**	Interface used by employees, managers, IT administrators and procurement staff to submit and manage requests.	**ServiceNow UI / Service Portal**
2	**Application Logic-1**	Handles laptop request submission and validation.	**ServiceNow Request Management**
3	**Application Logic-2**	Automates approval, procurement and status-update processes.	**ServiceNow Flow Designer**
4	**Application Logic-3**	Handles notifications and automated task creation.	**ServiceNow Flow Designer Actions & Notifications**
5	**Database**	Stores employee, laptop request, approval, procurement and order information.	**ServiceNow Tables / Glide Database**
6	**Cloud Database**	Platform-managed cloud data storage for ServiceNow applications.	**ServiceNow Cloud Database**
7	**File Storage**	Stores supporting documents or attachments related to procurement requests.	**ServiceNow Attachments / Platform Storage**
8	**External API-1**	Optional integration with an external inventory or asset management system.	**REST API**
9	**External API-2**	Optional integration with vendor/procurement systems for order processing.	**REST API**

10	**Machine Learning Model**	Not required for the core laptop procurement workflow.	**Not Applicable**
11	**Infrastructure (Server / Cloud)**	Hosts and runs the ServiceNow application and automated workflow.	**ServiceNow Cloud Platform**

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	The core solution is developed on the ServiceNow platform rather than using a separate open-source framework.	**ServiceNow Platform / Flow Designer**
2	**Security**	Role-based access ensures that employees, managers, IT administrators and procurement users can access only permitted functions.	**ServiceNow Roles, ACLs, Authentication**
3	**Scalable Architecture**	The cloud-based ServiceNow platform can support increasing numbers of users, requests and workflow activities.	**ServiceNow Cloud Platform**
4	**Availability**	The application is hosted on the ServiceNow cloud platform and can be accessed by authorized users through a web browser.	**ServiceNow Cloud Infrastructure**
5	**Performance**	Flow Designer automates repetitive processing and reduces manual processing time for requests, approvals, tasks and notifications.	**Flow Designer, Automated Workflows**

4. Technology Stack Summary

Layer	Technology Used
User Interface	ServiceNow UI / Service Portal
Application Platform	ServiceNow

Workflow Automation	Flow Designer
Request Management	ServiceNow Request Management
Approval	ServiceNow Approval Actions
Database	ServiceNow Tables
Notifications	ServiceNow Notifications
Access Control	Roles and ACLs
Integration	REST API
Hosting	ServiceNow Cloud Platform
Reporting	ServiceNow Dashboard / Reports
Machine Learning	Not Applicable

Overall Technology Flow

ServiceNow UI → Request Management → Flow Designer → Approval → Procurement Task → Notification → Order Tracking → ServiceNow Database

This architecture supports the project's main objective of **automating standard laptop procurement and reducing manual processing through ServiceNow Flow Designer**.